

module2 - DSA

Analysis of Loops in Algorithms

1. $O(1)$

$O(1)$ refers to constant time means that the running time of an algorithm remains constant and does not depend on the size of the input.

```
for i in range(1, 11):  
    # some  $O(1)$  expressions
```

2. Linear Time Complexity $O(n)$:

The Time Complexity is $O(n)$ if the loop variables are incremented/decremented by a constant amount

```
for i in range(1, n+1, c):  
    # some  $O(1)$  expressions
```

```
for i in range(n, 0, -c):  
    # some  $O(1)$  expressions
```

3. $O(n^2)$ - an algorithm whose performance is directly proportional to the squared size of the input data

```
for i in range(1, n+1, c):  
    for j in range(1, n+1, c):  
        # some  $O(1)$  expressions
```

4. Logarithmic Time Complexity $O(\log n)$:

The time Complexity of a loop is considered as $O(\log n)$ if the loop variables are divided/multiplied by a constant amount.

```
i = 1  
while(i <= n):
```

```
# some  $O(1)$  expressions
```

```
i = i * c
```